OOP with Java

lab sheet 04

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①
```
import java.util.scanner;

public class PrimeCheck{
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter a number: ");
        int num = sc.nextInt ();
        boolean isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= Math.sqrt (num); i++) {
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }
        if (isPrime)
            System.out.println (num + " is a prime number.");
        else
            System.out.println (num + " is not a prime number");

        sc.close ();
    }
}
```

②
```
import java.util.Scanner;

public class WordCount {
  public static void main (String [] args) {

    System.out.print ("Enter a sentence:");
    String sentence = sc.nextLine ();

    String [] words = sentence.trim().split ("\\s+");

    if (sentence.trim().isEmpty ()) {
      System.out.println ("The sentence has 0 words
    } else {
      System.out.println ("The sentence has " + words.length + "
                                                          w
    }
    sc.close();
  }
}
```

③
```
public class CountDown {
  public static void main (String [] args) {
    int number = 10;

    while (number >= 1) {
      System.out.println (number);
      number--;
    }
    System.out.println ("CountDown complete!");
  }
}
```

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④ import java.util.Scanner;

```
public class LeapYearCheck {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a year:");
    int year = sc.nextInt();

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
      System.out.println(year + " is a leap year.");
    } else {
      System.out.println(year + " is not a leap year.");
    }
    sc.close();
  }
}
```

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(5)

```
import java.util.Scanner;

public class CalculatorSwitch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter an operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        double result;

        switch (operator) {
            case '+':
                result = num1 + num2;
                System.out.println("Result: " + result);
                break;

            case '-':
                result = num1 - num2;
                System.out.println("Result: " + result);
                break;

            case '*':
                result = num1 * num2;
                System.out.println("Result: " + result);
                break;
```

```
        case '/':
            if (num2 != 0){
                result = num1/num2;
                System.out.println("Result: " + result);
            }else{
                System.out.println("Error: cannot divide by zero!
            }
            break;

        default:
            System.out.println("Invalid operator! Please use +,
                                *, or / ");
    }
    sc.close();
  }
}
```

⑥

```
import java.util.Scanner;
import java.util.array;

public class SortArray {
  public static void main (String [] args) {
    Scanner sc = new Scanner (System.in);
    int[] numbers = new int [10];

    System.out.println ("Enter 10 integers:");

    for (int i =0; i<10; i++) {
        System.out.print ("Number " + (i+1) + ": ");
        numbers[i] = sc.nextInt ();
    }

    Arrays.sort (numbers);

    System.out.println ("\nNumbers in ascending order:");
    for (int num: numbers) {
        System.out.print (num + " ");
    }
    sc.close();
  }
}
```

7.

```
import java.util.Scanner;

public class SumAndAverage{
  public static void main (String [] args){
    Scanner sc = new Scanner (System.in);

    double sum = 0;
    int count = 0;
    double number;

    System.out.println ("Enter numbers to sum. Enter 0
                        stop.");

    while (true) {
      System.out.print ("Enter a number; ");
      number = sc.nextDouble ();

      if (number == 0) {
        break;
      }

      sum += number;
      count++;
    }
    if (count > 0){
      double average=sum/count;
      System.out.print ln ("\nTotal sum: " + sum);
      System.out.println ("Average: " + average);
```

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```
else
} else {
System.out.println("No numbers were entered.")
}
sc.close();
}
}
```

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